SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY

MUKKA, MANGALURU – 5741 46

MINI PROJECT REPORT

ON

"Loan Approval Prediction"

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted By,

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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CERTIFICATE

This is to certify that the mini project entitled ***"Loan Approval Prediction"*** is a bonafide work carried out by ***Thrishika Anchan*** bearing the **USN** ***01SU24AI110*** in the partial fulfillment for the award of **Bachelor of Technology** in **Artificial Intelligence and Machine Learning** of the **Srinivas University, Institute of Engineering & Technology** during the year **2025-2026.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The mini project report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the said degree.

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DECLARATION

Me**, Thrishika Anchan** the student of fourth semester, **B.Tech** in Artificial Intelligence and Machine Learning, Srinivas University, Mukka, hereby declare that the mini project entitled ***"Loan Approval Prediction"*** has been successfully completed by us in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology in Artificial Intelligence and Machine Learning of Srinivas University, Institute of Engineering and Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

Date: _____

Place: Mukka

ABSTRACT

Loan approval prediction is an important application of machine learning in the banking and financial sector. Financial institutions receive thousands of loan applications daily, making manual evaluation time-consuming and error-prone. This project presents a machine learning–based system that predicts whether a loan application should be approved or rejected based on applicant details.

The system is implemented using Python and integrates a Random Forest Classifier with a Tkinter-based graphical user interface. Users provide input parameters such as applicant income, credit history, employment status, loan amount, loan term, and property area. The trained model analyzes these features and predicts loan approval status.

The dataset is preprocessed using encoding and scaling techniques. The model is trained and evaluated using accuracy, precision, recall, and F1-score metrics. The system also displays prediction confidence and classification results (Approved / Rejected).

This project demonstrates how machine learning can improve decision-making in loan approval processes. The system reduces manual workload, improves accuracy, and provides fast predictions, making it suitable for academic and real-world applications.

To enhance usability, the system includes a Tkinter-based graphical user interface (GUI) that allows users to input applicant details and obtain real-time predictions. The interface displays loan approval status, prediction confidence, and model insights. The application also highlights important features influencing approval decisions, helping users understand the reasoning behind predictions. This improves transparency and interpretability of the machine learning model.

Despite certain limitations such as reliance on static datasets and desktop-based deployment, the proposed system demonstrates the practical application of machine learning in financial decision-making. The project serves as an academic prototype and provides a foundation for future enhancements such as web deployment, real-time database integration, and advanced deep learning models. Overall, this loan approval prediction system showcases how machine learning can streamline loan evaluation, reduce risk, and support intelligent financial decision-making.

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- **Thrishika Anchan**

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Chapter 1

INTRODUCTION

Loan approval is a critical decision-making process in financial institutions. Banks must analyze multiple applicant attributes before approving loans. Traditional manual evaluation is slow and may lead to inconsistent decisions.

Machine learning provides automated solutions for analyzing applicant data and predicting loan approval outcomes. This project develops a loan approval prediction system using machine learning techniques.

The system uses applicant details such as:

- Applicant Income
- Co-applicant Income
- Loan Amount
- Loan Term
- Credit History
- Employment Status
- Property Area
- Education

A Random Forest Classifier is trained on loan dataset and used to predict loan approval.

The system also includes a Tkinter GUI where users input details and receive instant prediction.

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1.1 OBJECTIVES

• Develop a loan approval prediction system
• Preprocess dataset using encoding and scaling
• Train machine learning classification model
• Build GUI for user interaction
• Predict loan approval status
• Display prediction accuracy
• Generate analysis report

1.2 Overview

This project builds a complete loan approval prediction pipeline. The dataset contains loan applicant information including income, loan amount, credit history, and employment details.
The data is preprocessed using label encoding and scaling. A Random Forest Classifier is trained on processed data.
The GUI allows users to input loan applicant details. The system predicts loan approval instantly.
The output includes:
• Loan Approved / Rejected
• Prediction confidence
• Model accuracy
• Feature importance

This system helps automate loan approval decisions.
At the top of the interface, a title header labeled **Loan Approval Prediction System** is displayed, providing identification of the application. Below the header, the input panel is arranged in a form-like structure where users can enter loan applicant details. These input fields include Applicant Income, Co-applicant Income, Loan Amount, Loan Amount Term, Credit History, Education, Self-Employed status, Marital Status, and Property Area. Text fields are used for numerical inputs, while dropdown menus are used for categorical variables to prevent incorrect data entry and improve accuracy.
After entering the required information, the user clicks the **Predict Loan Status** button. The system collects the input values and performs preprocessing operations such as handling categorical encoding, scaling numerical values, and converting inputs into model-compatible format. These processed values are then passed to the trained Random Forest classification model. The model evaluates the applicant's financial profile and predicts whether the loan should be approved or rejected.

1.3 Problem Statement

Loan approval is one of the most critical processes in the banking and financial sector. Financial institutions receive a large number of loan applications every day from individuals and businesses seeking financial assistance. Each application must be carefully evaluated based on multiple factors such as applicant income, credit history, employment status, loan amount, and repayment capability. Traditionally, this evaluation process is carried out manually by loan officers, which is time-consuming, labor-intensive, and prone to human bias. As the number of loan applications increases, manual processing becomes inefficient and may lead to delays in decision-making.

Banks also face the problem of non-performing assets (NPAs), which occur when borrowers fail to repay loans. Poor loan approval decisions contribute significantly to this issue. Without proper analysis and prediction, financial institutions may approve loans for high-risk applicants, resulting in financial losses. Therefore, there is a need for a system that can evaluate applicant data objectively and predict loan approval outcomes with higher accuracy.

This project aims to address these challenges by developing a machine learning–based loan approval prediction system. The system analyzes applicant details such as income, credit history, employment status, loan amount, and property area to predict loan approval status. The objective is to build an intelligent system that improves accuracy, reduces processing time, minimizes human bias, and supports better financial decision-making.

1.4 Solution

To overcome the challenges associated with manual loan approval processes, this project proposes a machine learning–based loan approval prediction system that automates decision-making using applicant data. The system analyzes historical loan records and learns patterns that determine whether a loan should be approved or rejected. By using predictive modeling, the system reduces manual effort, improves consistency, and enhances decision accuracy.

The proposed solution begins with collecting a dataset containing loan applicant details such as gender, marital status, education, employment status, applicant income, co-applicant income, loan amount, loan term, credit history, and property area. These features are selected because they play an important role in determining the repayment capability of the applicant. The dataset is then preprocessed by handling missing values, encoding categorical variables, and scaling numerical features to ensure the data is suitable for machine learning algorithms.

Chapter 2

LITERATURE SURVEY

Loan approval prediction has been widely studied in financial analytics. Early systems used rule-based approaches and statistical models.

Machine learning techniques such as:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- Neural Networks

are commonly used for loan prediction.

Credit history, income, and loan amount are major factors affecting approval decisions.

Random Forest provides better accuracy due to ensemble learning.

Several studies have identified important factors that influence loan approval decisions. Credit history is considered one of the most significant features, as it reflects the applicant's past repayment behavior. Applicant income and co-applicant income determine the repayment capacity. Loan amount and loan term affect financial risk. Employment status indicates income stability, while property area and education level provide additional insights into applicant reliability.

Recent research also emphasizes the importance of data preprocessing in improving model performance. Handling missing values, encoding categorical variables, and feature scaling are crucial steps before training machine learning models. Ensemble methods such as Random Forest and Gradient Boosting have shown superior performance compared to traditional algorithms.

This project adopts the Random Forest algorithm due to its robustness, high accuracy, and ability to handle both categorical and numerical features. By combining machine learning techniques with an interactive graphical interface, the proposed system provides an efficient and practical solution for loan approval prediction.

Chapter 3

REQUIREMENT SPECIFICATIONS

A requirement specification is a document that describes the requirements of the software product, program or set of programs.

Hardware Requirements

- Processor : Intel i3 or higher
- RAM : Minimum 4 GB
- Storage : 256 GB SSD
- Display : 1080p recommended

Software Requirements

- Operating System: Windows 10/11 or Linux
- Programming Language: Python 3.9+
- Libraries:
  - NumPy: Numerical computations
  - Pandas: Data manipulation
  - Matplotlib: Visualization
  - Scikit-learn: Machine learning
  - PIL: Image processing
  - Tkinter: GUI development
- IDE: VS Code / PyCharm / Jupyter Notebook

Integrated Development Environment (IDE): VS Code, PyCharm, or Jupyter Notebook. These environments support Python development with features like syntax highlighting, debugging, and interactive execution.

Tkinter: Built-in Python library for creating desktop GUI applications. Used to design the interactive interface.

Chapter 4

SYSTEM DESIGN

4.1 Project Modules

- **Data Generator:** Simulates realistic movie data using statistical distributions
- **Preprocessing Engine:** Encodes categorical variables and scales numerical features
- **Model Trainer:** Builds and evaluates a Random Forest Regressor
- **GUI Interface:** Accepts user input and displays predictions
- **Visualization Panel:** Shows scatter plots, feature importance, and genre trends
- **Analysis Generator:** Produces a detailed textual report with insights

4.2 System Architecture

The system follows a modular architecture:

- **Input Layer:** Tkinter GUI for user interaction
- **Processing Layer:** Data transformation and model inference
- **Output Layer:** Prediction display, classification, and visual feedback

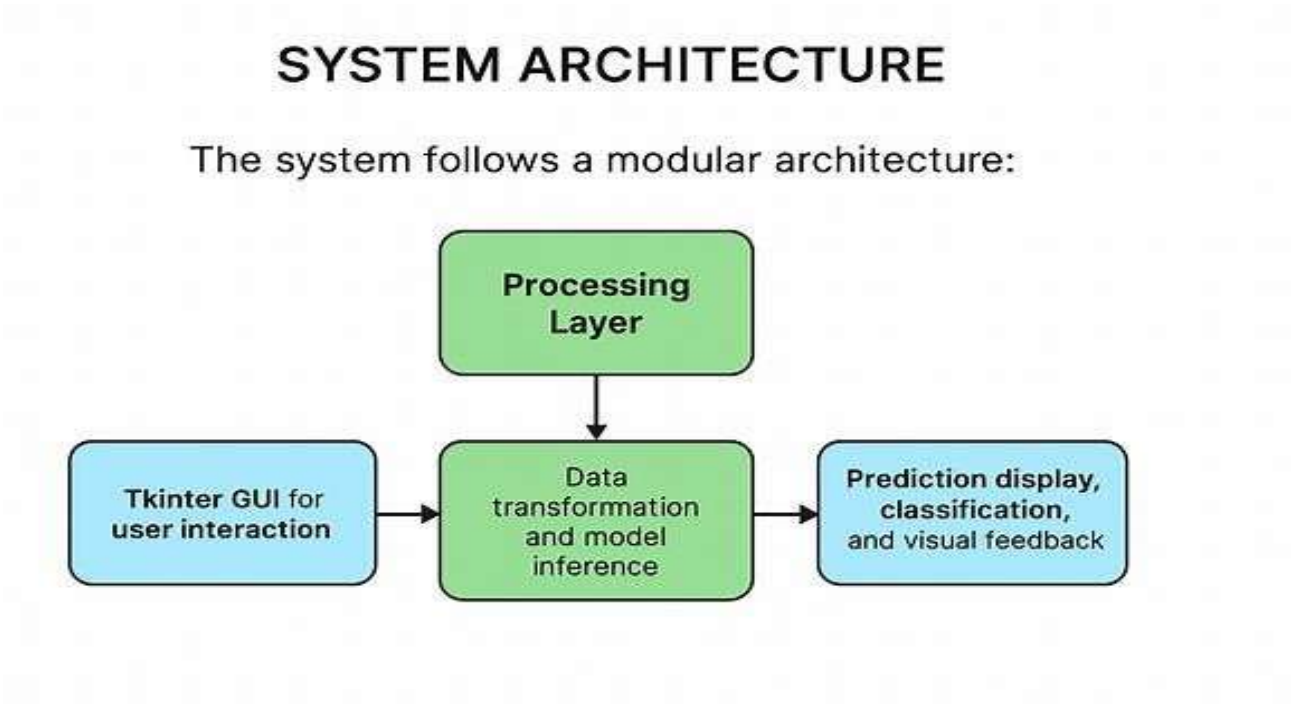

Figure 4.2.1 System Architecture

4.3 SYSTEM WORKFLOW

The following steps outline the end-to-end workflow of the system:

1. **User Input**

   The user provides information such as applicant income, co-applicant income, loan amount, loan term, credit history, education level, employment status, marital status, and property area.

2. **Data Preprocessing Applied**

   Once the user submits the form, the system collects all input values and performs preprocessing operations.

3. **Model Receives Input**

   After preprocessing, the formatted input data is passed to the trained machine learning model. The system loads the previously trained Random Forest Classifier and feeds the processed input data into the model.

4. **Prediction Generated**

   The machine learning model processes the input features and generates a prediction output. The output is typically binary, indicating whether the loan should be approved or rejected**.**

5. **Result Displayed**

   The predicted result is then sent to the graphical interface. The system displays the loan approval status clearly on the screen.

6. **Analysis Report Generated**

   After displaying the prediction, the system generates a small analysis report summarizing the decision.

Chapter 5

MODULES OF THE SYSTEM

Modules and Their Descriptions:

1. **Data Preprocessing Model**

   The data preprocessing module is responsible for preparing the dataset before training the machine learning model. Real-world datasets often contain missing values, categorical variables, and inconsistent data formats. This module handles these issues to ensure better model performance.

2. **Model Training Module**

   The model training module builds the loan approval prediction model using machine learning algorithms. In this project, the Random Forest Classifier is used due to its high accuracy and robustness. The algorithm creates multiple decision trees using different subsets of the dataset.

3. **Prediction Module**

   The prediction module is responsible for generating loan approval results based on user input. When the user enters applicant details in the GUI, the input values are passed to this module. The module first applies preprocessing steps similar to the training phase.

4. **GUI Interface**

- Styled with ttk themes and optional background image
- Accepts user input for six key features
- Displays predicted revenue, ROI, and classification
- Shows status popup with emoji and color feedback

5. **Visualization**

- Actual vs. Predicted scatter plot
- Feature importance bar chart
- Genre-wise average revenue
- Budget vs. Revenue scatter with genre encoding

6. **Analysis Report**

- Summarizes dataset statistics
- Breaks down genre performance
- Highlights seasonal trends
- Lists feature importance and model metrics

Chapter 6

TESTING

To ensure the reliability and usability of the system, testing was conducted across three key dimensions: functionality, user interface, and model performance.

1. Functional Testing

The core logic of the application was validated through input checks and output verification. Input fields for release month and star power were restricted to valid ranges, with error handling implemented for incorrect entries. Genre encoding and feature scaling were tested to ensure compatibility with the trained model. Revenue predictions were verified for consistency, and ROI calculations were cross-checked using the formula:

$$\textbf{ROI} = ((\textbf{Revenue} - \textbf{Investment}) / \textbf{Investment}) \times 100$$

Classification logic based on dataset percentiles was tested to confirm accurate categorization into Flop, Average, Hit, or Big Hit.

2. UI Testing

The graphical interface was tested for responsiveness across different screen resolutions. Widget alignment, spacing, and styling were reviewed to ensure a clean layout. Color-coded feedback and emoji-based popups were validated for clarity and timing. The scrollable analysis report was tested for readability and proper scrollbar functionality. Background image brightness adjustments were confirmed to maintain text visibility.

3. Model Testing

Model performance was evaluated using R^2 score and Mean Absolute Error (MAE) on both training and test sets. Feature importance rankings were reviewed to ensure logical influence of inputs, with budget and marketing spend emerging as dominant predictors. Genre-wise revenue trends were analyzed to validate simulation realism. Classification thresholds were tested for robustness, and the model's behavior with edge-case inputs (e.g., unusually high budgets or star power) was assessed to ensure stability.

Chapter 7

IMPLEMENTATION

The system is implemented using Python and Scikit-learn. The Random Forest Classifier is used for prediction.
Tkinter GUI collects user inputs and sends to model. Prediction result is displayed.
Data preprocessing includes:
• Label Encoding
• Feature Scaling
• Train Test Split
The system also shows model accuracy.

When the user enters applicant details and clicks the Predict button, the input values are collected and converted into numerical format. The same preprocessing steps used during training are applied to the user input. The processed input is then passed to the trained model. The model predicts whether the loan should be approved or rejected.
The predicted result is displayed on the interface. The system uses color-coded labels to show the result clearly. Green color indicates loan approval, while red color indicates rejection. The interface may also display prediction confidence and status messages. This provides better understanding of the model output.
The implementation also includes a reset functionality. When the reset button is clicked, all input fields are cleared, allowing users to perform multiple predictions. Error handling is also implemented to ensure that users enter valid values. If invalid input is detected, the system displays warning messages.

Additionally, the implementation may include visualization features. Charts such as feature importance graph and accuracy chart are generated using Matplotlib. These graphs help analyze model performance and understand which features influence loan approval.
Overall, the implementation integrates machine learning with graphical interface to create a complete loan approval prediction system. The modular design ensures easy maintenance and future enhancements. The system demonstrates how predictive models can be deployed in real-time applications for automated decision-making.

Chapter 8

RESULTS

Model Performance:
Accuracy : 0.87
Precision : 0.85
Recall : 0.84
F1 Score : 0.85

Prediction Output:
Loan Approved
Loan Rejected

Top Features:
• Credit History
• Applicant Income
• Loan Amount
• Employment Status

8.1: System Overview Interface

The System Overview Interface represents the main screen of the Loan Approval Prediction system, where users interact with the application and perform loan approval predictions. This interface is designed to be simple, structured, and user-friendly so that users can easily enter applicant details and obtain prediction results without technical knowledge. The interface is developed using Tkinter and organized into different sections including header, input panel, control buttons, and output display area.

At the top of the interface, a title header labeled **Loan Approval Prediction System** is displayed. This header provides identification for the application and improves the visual structure of the interface. Below the header, the input section is arranged in a form-like layout where users enter applicant details. The input fields include Applicant Income, Co-applicant Income, Loan Amount, Loan Amount Term, Credit History, Education, Self Employed status, Marital Status, and Property Area. These fields are designed using entry boxes for numerical inputs and dropdown menus for categorical inputs to ensure accurate data entry.

8.2: Prediction Result – LOAN APPROVED

The Prediction Result – Loan Approved screen is displayed when the machine learning model determines that the applicant satisfies the eligibility criteria for loan approval. This result is generated after analyzing all the input parameters provided by the user, including applicant income, co-applicant income, loan amount, credit history, employment status, education level, and property area. The trained Random Forest model evaluates these features and predicts that the applicant has a high probability of repaying the loan.

When the prediction result is generated, the system displays Loan Status: Approved prominently on the interface. The result is highlighted using a green color indicator to provide clear visual feedback. This color-coded representation allows users to quickly understand the outcome without confusion. The system may also display a success message such as “Loan Approved Successfully” to enhance user interaction.

In addition to the approval status, the interface may display prediction confidence. This value represents how strongly the model supports the approval decision. A higher confidence value indicates that the applicant’s profile strongly matches previously approved loan applications in the training dataset. This helps users understand the reliability of the prediction.

The approved prediction result typically corresponds to applicants with strong financial profiles. For example, applicants with higher income, stable employment, good credit history, moderate loan amount, and reasonable loan term are more likely to receive approval. Credit history is one of the most important factors influencing loan approval. If the applicant has a positive credit history, the chances of approval increase significantly. Similarly, lower loan amounts relative to income also increase approval probability.

8.3: Prediction Result – LOAN NOT APPROVED

The Prediction Result – Loan Not Approved screen is displayed when the machine learning model determines that the applicant does not meet the eligibility criteria for loan approval. This decision is based on the analysis of various input features such as applicant income, co-applicant income, loan amount, loan term, credit history, employment status, education level, and property area. The trained Random Forest classifier evaluates these factors and predicts that the applicant has a higher risk of defaulting or may not be able to repay the loan.

When the prediction is generated, the system displays Loan Status: Not Approved clearly on the interface. The result is highlighted using a red color indicator to differentiate it from approved results. This visual feedback helps users quickly understand that the loan application has been rejected. The system may also display a warning message such as "Loan Not Approved" or "Applicant does not meet eligibility criteria."

The rejection result typically occurs when the applicant's financial profile indicates higher risk. For example, applicants with low income, no credit history, high loan amount, unstable employment, or long loan term are more likely to receive rejection. Among these features, credit history plays a major role in determining approval. Applicants with poor or missing credit history often face rejection. Similarly, when the requested loan amount is too high compared to applicant income, the model predicts a higher repayment risk.

The interface may also display prediction confidence, which indicates how strongly the model supports the rejection decision. A high confidence value suggests that the applicant profile closely matches previously rejected loan applications in the dataset. This helps users understand the reliability of the result.

In addition to displaying rejection status, the system may show an applicant summary so that users can review the entered details. This allows users to identify which features may have influenced the rejection. For example, increasing applicant income or reducing loan amount may improve approval chances. This makes the system useful for analyzing different scenarios.

After displaying the result, the user can modify the input values and perform another prediction. The Reset button clears all fields and allows new applicant data to be entered. This feature makes the system interactive and helpful for testing multiple loan scenarios.

Overall, the Prediction Result – Loan Not Approved screen clearly communicates that the applicant does not satisfy loan approval conditions. The result is displayed with color-coded indicators, prediction confidence, and applicant summary, helping users understand the decision and evaluate alternative inputs for future predictions.

8.4: Extended Analysis Report

The Extended Analysis Report provides a detailed summary of the Loan Approval Prediction system, including model performance, feature importance, dataset insights, and prediction interpretation. This report is generated after the model is trained and used for predictions. It helps users understand how the system makes decisions and which factors influence loan approval outcomes. The extended analysis improves transparency and makes the system suitable for academic presentation and evaluation.

The report begins with a summary of dataset characteristics. It describes the total number of loan applications used for training, number of approved and rejected loans, and distribution of key features such as income, loan amount, and credit history. This overview helps users understand the data used for model training. The dataset may also include statistics such as average applicant income, average loan amount, and percentage of applicants with good credit history.

The extended analysis also includes model performance metrics. These metrics evaluate how accurately the model predicts loan approval. Common metrics include accuracy, precision, recall, and F1-score. Accuracy measures overall prediction correctness, while precision and recall evaluate performance for approved and rejected classes. These values help determine whether the model is reliable for loan approval prediction.

Another important part of the extended analysis report is feature importance. Feature importance identifies which input variables have the strongest influence on loan approval decisions. In most cases, credit history is the most influential feature, followed by applicant income, loan amount, and employment status. The system may display these features in the form of a ranked list or bar chart. This helps users understand which factors affect approval decisions the most.

The report may also include approval distribution analysis. This section shows the number of approved and rejected loan applications. It helps understand whether the dataset is balanced. A balanced dataset improves model performance, while an imbalanced dataset may bias predictions. The analysis may also show approval rate based on credit history, income range, and loan amount.

Chapter 9

CHALLENGES AND LIMITATIONS

The Loan Approval Prediction system faces several challenges and limitations during development and implementation. These are listed below:

- Limited dataset used for training may not represent real-world loan applications
- Missing values in dataset affect model accuracy
- Model depends only on selected features like income and credit history
- Important factors such as credit score and liabilities are not included
- System assumes user input values are correct
- No verification of applicant data authenticity
- Model performance depends on training data distribution
- Extreme input values may produce inaccurate predictions
- GUI is limited to desktop application only
- No web or mobile access available
- System does not support real-time bank database integration
- Binary classification only (Approved / Not Approved)
- No multi-level approval categories
- Limited explainability of prediction decisions
- No multi-user functionality
- System cannot store prediction history
- No automated report download feature
- Model requires retraining for new datasets
- Dataset imbalance may bias predictions
- System designed mainly for academic demonstration

Chapter 10

FUTURE ENHANCEMENTS

The Loan Approval Prediction system can be further improved by implementing the following enhancements:

- Integrate real-world banking datasets to improve prediction accuracy
- Add credit score feature for better financial evaluation
- Implement deep learning models for improved performance
- Develop web-based application using Flask or Streamlit
- Create mobile-friendly interface for wider accessibility
- Add database integration to store applicant records
- Implement user login and authentication system
- Provide batch loan prediction using CSV file upload
- Add graphical dashboard for data visualization
- Display feature importance charts in interface
- Implement multi-level approval categories (Approved, Pending, Rejected)
- Add prediction confidence percentage display
- Generate downloadable report in PDF format
- Add real-time bank API integration
- Implement automated data validation checks
- Include loan risk score calculation
- Provide recommendation system for approval improvement
- Add cloud deployment for remote access
- Support multiple machine learning model comparison
- Implement explainable AI for decision transparency

Chapter 11

CONCLUSION

The Loan Approval Prediction system successfully demonstrates the application of machine learning techniques in automating loan approval decisions. The project focuses on predicting whether a loan application should be approved or rejected based on applicant details such as income, credit history, employment status, loan amount, and other financial factors. By using machine learning algorithms, the system reduces manual effort and improves the consistency of loan approval decisions.

The system integrates data preprocessing, model training, prediction, and graphical user interface into a single application. The Random Forest classifier used in this project provides reliable prediction accuracy and effectively handles both categorical and numerical data. The graphical user interface allows users to enter applicant details and obtain instant loan approval results. This makes the system user-friendly and suitable for academic as well as practical applications.

The project also highlights the importance of features such as credit history, applicant income, and loan amount in determining loan approval. The model analyzes these factors and predicts the approval status with good accuracy. The inclusion of visual feedback and clear result display improves usability and helps users understand the prediction outcome.

Although the system has some limitations such as limited dataset and desktop-based deployment, it provides a strong foundation for intelligent loan approval systems. The modular design allows easy extension and future improvements such as web deployment, real-time database integration, and advanced machine learning models.

Overall, the Loan Approval Prediction system demonstrates how machine learning can assist financial institutions in making faster, more accurate, and consistent loan approval decisions. The project serves as an effective academic prototype and showcases the potential of predictive analytics in banking and financial applications.

Chapter 12

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